Name: ______________________ Date: Day ______________ Period: ____________

Conversion Practice

Volume&Capacity

2 liters = ____________ milliliters

______ = ______

3 quarts = 6 pints

$$\times 2\left(\frac{3\,qt}{x\,pt} = \frac{1\,qt}{2\,pt}\right)\times 2$$

SF: x2

Length

75 centimeters = 0.75 meters

$$\div 100\left(\frac{75cm}{x\,m} = \frac{100cm}{1\,m}\right)\div 100$$

SF: $\times\frac{1}{100}$

12 feet = 4 yards

$$\div 3\left(\frac{12ft}{x\,yd} = \frac{3ft}{1yd}\right)\div 3$$

SF: $\times\frac{1}{3}$

Weight & Mass

6 pounds = 96 ounces

$$\times 16\left(\frac{6\,lb}{x\ oz} = \frac{1\,lb}{16\,oz}\right)\times 16$$

SF: x16

2,500 grams = 2.5 kilograms

$$\div 1000\left(\frac{2{,}500\,g}{x\,kg} = \frac{1{,}000\,g}{1\,Kg}\right)\div 1000$$

SF: $\times\frac{1}{1000}$

Snake Method

What if the STAAR chart does not give you a direct conversion? Use the "snake method"

3 gallons = 3x16 = 48 cups

SF: 4x2x2 = x16

Customary

1 gallon (gal) = 4 quarts (qt)

1 quart (qt) = 2 pints (pt)

1 pint (pt) = 2 cups (c)

$$\times 4\left(\frac{3\,gal}{x} = \frac{1\,gal}{4\,qt}\right)\times 4$$

x4

$$\times 2\left(\frac{12\,qt}{x} = \frac{1\,qt}{2\,pt}\right)\times 2$$

x2

$$\times 2\left(\frac{24\,pt}{x} = \frac{1\,pt}{2\,cup}\right)\times 2$$

x2

Snake Method:

2 miles = 10,560 feet

SF: x1760 x 3 = x5280

Customary

1 mile (mi) = 1,760 yards (yd)

1 yard (yd) = 3 feet (ft)

1 foot (ft) = 12 inches (in.)

$$\frac{2\ mi}{x} = \frac{1\,mi}{1760\,yd}$$

X1760

$$\times 3\left(\frac{3520\,yd}{x\ ft} = \frac{1\,yd}{3\,ft}\right)\times 3$$

x3

NAME ______________________ DATE ______________ PERIOD ________

Problem-Solving Practice: *Convert Measurement Units*

1. The average weight of a baby at birth is 7 pounds. How many ounces is the average weight of a baby?

$$\frac{7 \text{ lb}}{x \text{ oz}} = \frac{1 \text{ lb}}{16 \text{ oz}} \quad \times 7$$

$16 \times 7 = 112$

$\boxed{x = 112 \text{ ounces}}$

2. The height of Niagara Falls is 56 meters. How many kilometers high is Niagara Falls?

$$\div 1{,}000 \quad \frac{56 \text{ m}}{x \text{ km}} = \frac{1{,}000 \text{ m}}{1 \text{ km}} \quad \div 1{,}000$$

$\boxed{x = 0.056 \text{ km}}$

3. The gasoline tank of a minivan holds 18 gallons. How many quarts can the tank hold?

$$\times 4 \quad \frac{18 \text{ gal}}{x \text{ qt}} = \frac{1 \text{ gal}}{4 \text{ qt}} \quad \times 4$$

$18 \times 4 = 72$

$\boxed{x = 72 \text{ quarts}}$

4. Cell phones can weigh as little as 56 grams. How many kilograms can the cell phone weigh?

$$\div 1000 \quad \frac{56 \text{ g}}{x \text{ kg}} = \frac{1{,}000 \text{ g}}{1 \text{ kg}} \quad \div 1000$$

$\boxed{x = 0.056 \text{ kg}}$

5. A recipe for ice cream calls for 1.656 liters of milk. How many milliliters of milk are in the recipe?

$$\times 1{,}000 \quad \frac{1.656 \text{ L}}{x \text{ mL}} = \frac{1 \text{ L}}{1{,}000 \text{ mL}} \quad \times 1{,}000$$

$\boxed{x = 1{,}656 \text{ mL}}$

6. The Statue of Liberty weighs 450,000 pounds. How many tons does the statue weigh?

$$\div 2{,}000 \quad \frac{450{,}000 \text{ lb}}{x \text{ tons}} = \frac{2{,}000 \text{ lb}}{1 \text{ ton}} \quad \div 2{,}000$$

$\boxed{x = 225 \text{ tons}}$

7. The Ted Williams Tunnel under Boston Harbor is 2.57 kilometers long. How many meters is the length of the tunnel?

$$\times 1{,}000 \quad \frac{2.57 \text{ km}}{x \text{ m}} = \frac{1 \text{ km}}{1{,}000 \text{ m}} \quad \times 1{,}000$$

$\boxed{x = 2{,}570 \text{ m}}$

8. The United States exports over 200 billion pounds of coal. How many tons does the United States export?

$$\div 2{,}000 \quad \frac{200{,}000{,}000{,}000 \text{ lb}}{x \text{ tons}} = \frac{2{,}000 \text{ lb}}{1 \text{ ton}} \quad \div 2{,}000$$

$\boxed{x = 100{,}000{,}000 \text{ tons}}$